

Weekly Report

iOS Application Development

Date: April 14 – April 21, 2026
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Project: Employ210 – Vocational Training Platform

1. Task Evaluation Screen

A comprehensive Task Evaluation interface was developed this week, enabling trainers to assess trainee performance on Hierarchical Task Analysis (HTA) programs. This feature is central to the application's mission of providing structured vocational training and measurable progress tracking for individuals with developmental disabilities.

Key Features Implemented:

4-State Evaluation Cycle: Each step in an HTA program can be evaluated by tapping to cycle through four distinct states. This provides nuanced feedback beyond simple pass/fail assessment:

State	Color	Meaning
Not Evaluated	Grey	Step has not been assessed yet
Pass	Green	Trainee completed the step independently
Needs Prompting	Orange	Trainee required assistance or verbal cues
Not Completed	Red	Trainee was unable to complete the step

Stopwatch Timer: A built-in timer counts upward to track how long a trainee takes to complete a task. Trainers can Start, Pause, and Reset the timer as needed. The elapsed time is automatically saved with the evaluation results for performance analysis and progress tracking over multiple sessions.

Live Statistics Bar: A real-time statistics display shows the count of steps in each evaluation state (Pass, Needs Prompting, Not Completed, Pending), giving trainers an at-a-glance view of trainee performance during the evaluation session.

Visual Feedback & Haptics: Each step card displays color-coded indicators and status badges for clear visual communication. Haptic feedback is triggered on each tap for tactile confirmation, improving the user experience during hands-on training sessions.

Evaluation Storage: An EvaluationStore system persists all evaluation results locally, including elapsed time, individual step results, timestamps, and summary statistics. This data enables longitudinal progress tracking and reporting capabilities.

2. Record & Analyze Feature

The Record & Analyze feature enables trainers to capture video recordings of trainees performing tasks. These recordings serve multiple purposes: real-time observation, post-session review, documentation for progress reports, and future AI-powered behavioral analysis.

Implementation Details:

Camera Integration: Full camera preview functionality using Apple's AVFoundation framework. The interface includes intuitive controls to start and stop recording, with a visual recording indicator (pulsing red dot) when actively capturing video.

AWS S3 Cloud Storage: Recorded videos are uploaded to Amazon S3 with an organized folder structure for easy retrieval and management. The path structure follows the format: *recordings/{trainerId}/{traineeId}/recording_{timestamp}.mov*, enabling organization by trainer and trainee relationships.

Upload Progress Interface: A circular progress indicator displays the upload percentage in real-time, keeping users informed during the upload process. Both success and error states are handled with appropriate user-facing alerts and messaging.

Metadata Management: Recording metadata including S3 storage keys, timestamps, and trainee/trainer identifiers are saved locally for quick reference and offline access without requiring network calls.

Current Status:

The recording integration is nearly complete. Core functionality has been validated in the iOS Simulator environment. Final work involves resolving AWS Cognito credential configuration for physical device deployment. Enhanced error logging has been implemented to expedite debugging and resolution of the remaining issues.

3. Collaboration & Meetings

Met with Katie this week to discuss ongoing work on the SNCC project and upcoming features for the Employ210 application. Key discussion topics included:

- New page designs and user interface improvements for the Employ210 application
- User experience enhancements based on trainer feedback
- Potential integration points between SNCC behavioral analysis capabilities and Employ210's training evaluation features
- Shared component architecture opportunities between both projects

Upcoming Tasks

Priority	Task	Target
High	Resolve AWS S3 upload configuration for physical devices	This week
High	Implement new pages discussed with Katie	Next week
Medium	Begin AI video analysis integration for recorded sessions	Ongoing
Medium	Add progress dashboard for trainee performance tracking	Upcoming